

exp 1.py

```
1 import pandas as pd
2 data = pd.DataFrame({
3     "Sky": ["Sunny", "Sunny", "Rainy", "Sunny"],
4     "AirTemp": ["Warm", "Warm", "Cold", "Warm"],
5     "Humidity": ["Normal", "High", "High", "High"],
6     "Wind": ["Strong", "Strong", "Strong", "Strong"],
7     "Water": ["Warm", "Warm", "Warm", "Cool"],
8     "Forecast": ["Same", "Same", "Change", "Change"],
9     "EnjoySport": ["Yes", "Yes", "No", "Yes"]
10 })
11 concepts = data.iloc[:, :-1].values
12 target = data.iloc[:, -1].values
13 hypothesis = None
14 for i in range(len(target)):
15     if target[i] == "Yes":
16         if hypothesis is None:
17             hypothesis = concepts[i].copy()
18         else:
19             for j in range(len(hypothesis)):
20                 if hypothesis[j] != concepts[i][j]:
21                     hypothesis[j] = "?"
22
23 print("Most Specific Hypothesis:")
24 print(hypothesis)
```

Run exp 1



```
/Users/moshika/PycharmProjects/PythonProject/.venv/bin/python /Users/moshika/PycharmProjects/PythonProject/exp 1.py
```

```
Most Specific Hypothesis:
```

```
['Sunny' 'Warm' '?' 'Strong' '?' '?']
```

```
Process finished with exit code 0
```